MICCAI 2026 Challenge • Clinical NLP • Biomedical Knowledge Representation

CohortX

Extracting Executable Cohort Definitions for Medical Imaging Research

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MICCAI2026

RESEARCH QUESTION

How can free-text cohort eligibility criteria be identified, structured, and semantically represented so that they become reproducible, comparable, and executable over imaging and integrated clinical data?

BACKGROUND & CHALLENGE GOAL

Clinical cohort eligibility is usually described in unstructured free text within scholarly articles and trial registries. This makes patient selection difficult to operationalise for large-scale medical image analysis and clinical translation.

CohortX targets machine-interpretable outputs that capture **logical, demographic, and temporal constraints** and enable executable cohort queries over imaging repositories and integrated clinical data.

FROM NARRATIVE TO QUERY

01 TEXT	02 CRITERIA	03 TRIPLES	04 CODES
Articles + registries	Extraction	Structure	ICD-10-CM + ontologies

Endpoint: executable cohort queries linking narrative evidence to imaging and clinical repositories.

METRICS

T1: number similarity, BERT cosine, FM3S ontology similarity, polarity consistency.
T2: FM3S semantic similarity, polarity consistency.
T3: F1-score.

Data Engineering and Semantics
هندسة البيانات و دلالاتها

CHALLENGE TASKS

T1 — COHORT CRITERIA EXTRACTION

Find cohort eligibility criteria from clinical trial registries in full-text scholarly publications.

Example: "Age ≥ 18 and ECOG ≤ 2" → aged 18 or more and having an ECOG less or equal to 2

Data: 416 training · 300 validation · 200 test

T2 — CRITERIA → TRIPLES

Convert extracted criteria into subject–predicate–object triples that preserve meaning and structure.

Example: "Age ≥ 18" → (Patient, hasAge, ≥18)

Data: 100 training · 25 validation · 25 test

T3 — CONDITION → ICD-10-CM

Map medical condition names to comprehensive ICD-10-CM code sets, including synonyms, subtypes, and variants.

Example: "Type 2 diabetes" → E11.9 and related codes

Data: 5 sample training cases · 23 test cases

PARTICIPATION SNAPSHOT

Task	Teams	Submissions	Validated
T1	58	1,860	3
T2	45	1,794	9
T3	123	5,613	6

123 T3 teams — largest field

9 T2 validated — highest count

PARTICIPATION & SCHEDULE

- **APR 1** Training data release; submissions begin.
- **APR 30** Registration deadline.
- **JUL 15** Submissions close (Kaggle).
- **AUG 1** Results announced.
- **OCT 1** Challenge Showcase at the **Leicester Room**, MICCAI 2026.

Prizes per task: 1st $120 · 2nd $100 · 3rd $80. Top 10 teams are offered co-authorship; source code shared under managed access.

FUTURE DIRECTION

Semantic Data Modeling (originally planned as T4) extends triples to full knowledge graphs using biomedical ontologies and Semantic Web standards.

FHIR · RDF · SPARQL · OWL · SNOMED International

Goal: executable cohort queries over EHR-linked imaging repositories.

EXPECTED IMPACT

REPRODUCIBLE RESEARCH
Explicit, inspectable eligibility logic.

COHORT DISCOVERY
Structured criteria enable search and comparison.

CLINICAL INTEROPERABILITY
Standard codes connect narrative evidence with biomedical data.

DOWNSTREAM AI
Supports evidence synthesis, cohort analytics, and patient–trial matching.

PARTICIPATION BY AFFILIATION

TASK 1
58 teams · 1,860 submissions · 3 validated

TOP COUNTRIES	
India	10
United States	8
China	5

CONTINENTS	COUNTRIES / PARTICIPANTS
Africa	4 / 7
Asia	12 / 30
Europe	7 / 11
N. America	2 / 9
Oceania	1 / 1

TASK 2
45 teams · 1,794 submissions · 9 validated

TOP COUNTRIES	
India	10
China	6
United States	5

CONTINENTS	COUNTRIES / PARTICIPANTS
Africa	4 / 5
Asia	11 / 28
Europe	4 / 5
N. America	2 / 6
Oceania	1 / 1

TASK 3
123 teams · 5,613 submissions · 6 validated

TOP COUNTRIES	
China	59
India	13
United States	10

CONTINENTS	COUNTRIES / PARTICIPANTS
Africa	5 / 10
Asia	13 / 93
Europe	7 / 8
N. America	2 / 11
S. America	1 / 1

SHOWCASE UPDATE

October 1, 2026 · Leicester Room
Challenge Showcase at MICCAI 2026.

COHORTX
Website: csisc.github.io/CohortX-MICCAI/

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#CohortX

ABOUT THIS POSTER
Based on the official CohortX challenge proposal for MICCAI 2026. Participation statistics are supplied by the challenge organizers.

CohortX Task 1

Extracting Cohort Selection Criteria from Free-Texts

BACKGROUND & AIM

Why it matters

Cohort-definition information is often buried in unstructured biomedical text, limiting scalable cohort selection, reproducibility, and direct querying of EHRs.

Task 1 objective

Extract six cohort fields from full-text articles, with registry-derived and paraphrased ClinicalTrials.gov records serving as the reference.

TASK & DATA

Six target fields

- Conditions
- Study type
- Sex
- Minimum age
- Maximum age
- Eligibility criteria

Dataset

416 labeled training records; 500 test records (300 public + 200 private).

EVALUATION

Official composite

Equal-weight arithmetic mean of the six field components.

Eligibility: FM3S

Section-aware similarity using noun, verb, and common-word-order features with WordNet semantics.

Robustness

2,000 bootstrap samples and 5,000 paired sign-flip permutations.

KEY METRIC CAVEATS

- Age scoring is unit-blind: “18 Years” and “18 Months” receive full numeric credit.
- Absolute word positions in FM3S can be fragile for long paraphrased text.
- Holistic eligibility similarity is not criterion-level precision, recall, or F1.

PARTICIPATING SYSTEMS

System A — Alan T. Andrea

Alan T. Andrea, OptiMLX Inc., United States of America

FLAN-T5 for conditions; TF-IDF/logistic regression for study type & sex; defaults for age; BioBERT weak supervision + classifier for eligibility.

System B — NTUA

Kyriaki Kolpetinou, Nikolas Robotis, George K. Matsopoulos, National Technical University of Athens, Greece

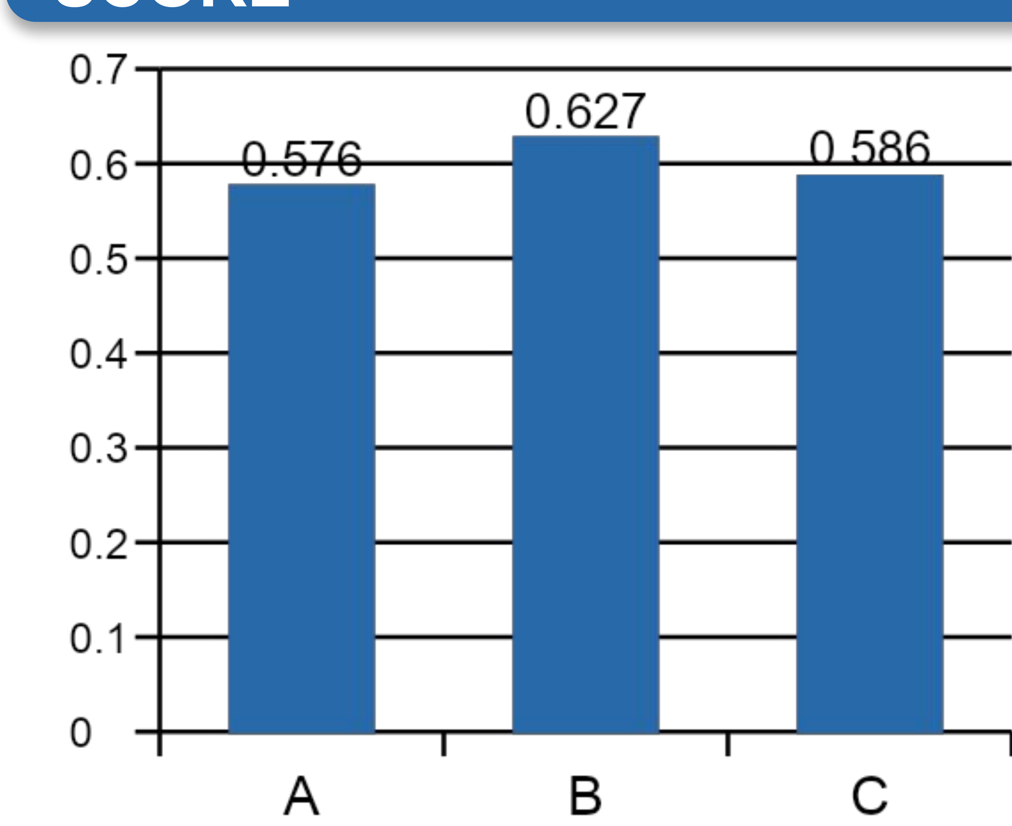
Qwen2.5-3B RAG few-shot for conditions; PubMedBERT classifier; regex sex rules; BART + RAFT for eligibility.

System C — IUCompPath

Sanket Kachole, Spyridon Bakas, Indiana University, United States of America

Hybrid passage ranking with BioBERT embeddings plus structural/lexical features; high-precision rules for other fields.

OFFICIAL COMPOSITE SCORE



System B ranked first: 0.627

B was significantly higher than A and C ($p < 0.001$); C vs A was not significant.

FIELD-SPECIFIC RESULTS

Field	A	B	C
Conditions	.365	.436	.306
Study type	.876	.870	.857
Sex	.876	.895	.904
Min age	.656	.676	.582
Max age	.502	.504	.422
Eligibility	.182	.382	.446

MAIN FINDINGS

1 System B leads overall

Composite 0.627, with the ranking unchanged under eligibility-heavy reweighting.

2 Eligibility is the central challenge

System C leads eligibility (0.446), but its longer outputs do not establish criterion-level correctness.

3 Conditions are highly discriminative

System B scores 0.436 vs 0.365 (A) and 0.306 (C).

PUBLIC → PRIVATE SHIFT

Composite scores

A: 0.587 → 0.560
B: 0.637 → 0.612
C: 0.596 → 0.571

Distribution differences

Eligibility length: 1,166 → 765 chars; maximum-age numeric values: 46.0% → 55.5%.

ELIGIBILITY OUTPUT PROFILE

Predicted mean length

A 115 chars • B 219 chars • C 1,917 chars

Lexical unit-overlap proxy

A 0.013 • B 0.054 • C 0.190

Interpretation

Overlap is exploratory—not validated criterion recovery.

CONCLUSIONS & FUTURE WORK

- Automated cohort-definition extraction is feasible under CPU/offline constraints.
- Benchmark scores should not be equated with clinically reliable cohort extraction.
- Next: criterion-level P/R/F1, provenance spans, and structured entities, negation, temporal and logical relations.

KEY REFERENCES

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Source: CohortX Task 1 on Kaggle • **Funding:** Saudi NIH PRI01-2401-KAU19-46641119 • **Contact:** ahalzhrani@kau.edu.sa | csisc.github.io/CohortX-MICCAI

CohortX Task 2

Structuring Cohort Eligibility Criteria as Semantic Triples

45	1,794	100 + 50	9	0.80	0.78–0.80
participants	submissions	train + test criteria	validated systems	top official score	top-4 score band

BACKGROUND & OBJECTIVE

The problem

- Free-text criteria are expressive but hard to compute, screen, and reproduce.
- ClinicalTrials.gov has 100,000s+ trials with unstructured criteria; prior work often jumps to OMOP, SQL, or bespoke schemas.

Task

Convert each free-text eligibility criterion into (subject, predicate, object) triples.

patient → has confirmed diagnosis → type 2 diabetes

Reference construction

ClinicalTrials.gov API
↓
Radiology trials
X-ray • CT • MRI
↓
GPT-5.2 triple generation
reification permitted
↓
2-annotator validation

Output format: em-dash separators • blank line between triples • no ontology binding

METHODS & EVALUATION

Challenge timeline

1 Apr–15 Jul	**Competition**	45 participants • 1,794 submissions
15–31 Jul	**Code review**	20 invited • 9 validated
1 Aug	**Final results**	Rankings published

FM3S + Hungarian matching

$$\mathrm{sim}(t_1,t_2) = [\mathrm{FM3S}(s_1,s_2) + \mathrm{FM3S}(r_1,r_2) + \mathrm{FM3S}(o_1,o_2)] / 3$$

Sim_N	Sim_V	CWO
WordNet-grounded noun similarity	Verb similarity	Content-word-order
Best-matching Lin similarity	Tense-sensitive matching	Shared word / bigram order

Optimal bipartite alignment

Gold(Pred) = argmax(Pred, G) for G in Reference

Unmatched triples = 0 • Row score = mean similarity over gold triples • Challenge score = mean row score

Why this representation?

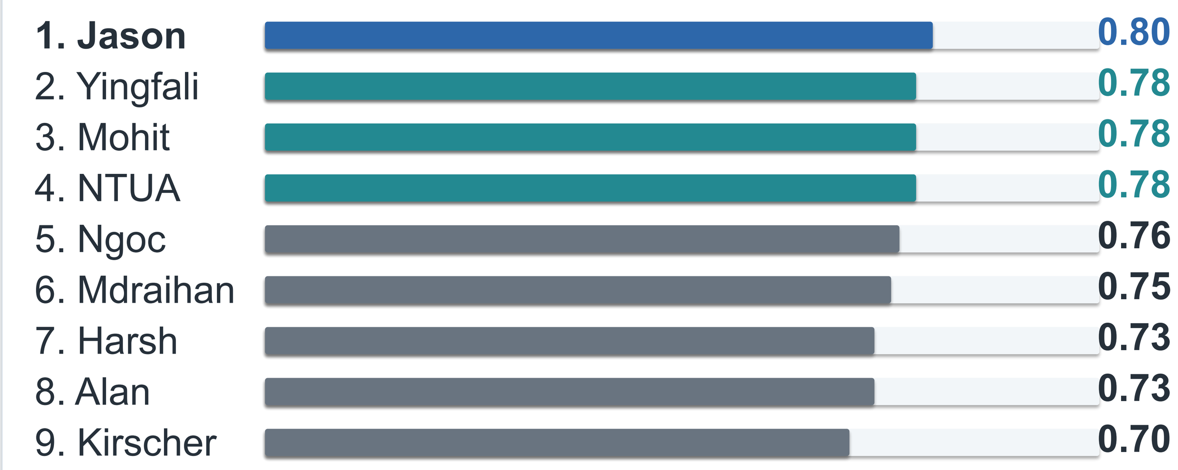
• Order-independent scoring mirrors unordered RDF graph semantics.

• Synonym tolerance avoids brittle exact-string matching.

• Extraction is evaluated before ontology / database binding.

RESULTS

Full-set leaderboard



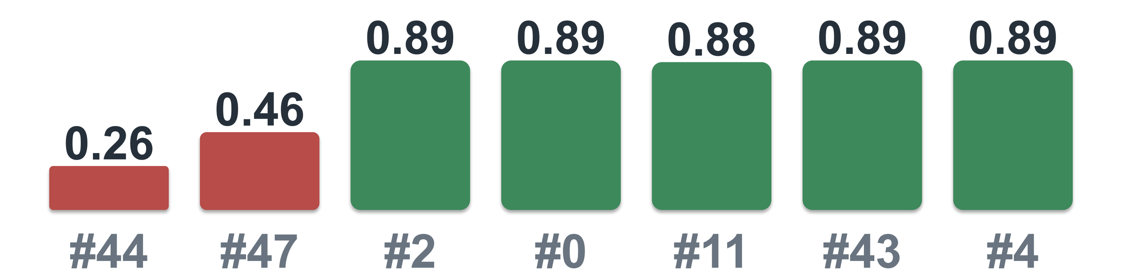
Solution upper bound: 1.00 official | raw mean 0.9752

Key finding: top 4 effectively tied

0.78–0.80

95% bootstrap CI overlap (±0.04–0.05). Wilcoxon tests found no significant difference between Jason and the other top-three systems (p > 0.05).

Difficulty varies sharply by record



Hardest: Record 44 = 0.257; Record 47 = 0.460 | Easiest records > 0.88

No single system dominates

8 / 50 Top systems each won 8 records. Cross-record performance was weakly or negatively correlated (mean Spearman $\rho \approx -0.12$).

CONCLUSION

CohortX Task 2 shows that converting eligibility criteria to semantic triples is feasible and benchmarkable.

Top systems: 0.78–0.80 • RDF/Turtle as next layer

LIMITATIONS

• LLM-generated reference standard (GPT-5.2); review extent is not fully documented.

• WordNet underperforms for acronyms, drug names, and proper nouns.

• Multiple valid decompositions may exist; only 9/20 invited systems submitted validated code.

• ClinicalTrials.gov-only source limits generalization.

IMPLICATIONS FOR BIOMEDICAL INFORMATICS

• Triple layer cleanly separates extraction from ontology binding.

• Triples map directly to RDF graph structure and are Turtle-ready once vocabularies are declared.

• Model-agnostic representation enables comparison across architectures.

FUNDING

Saudi National Institute of Health (Saudi NIH) Grant PRI01-2401-KAU19-46641119

CONTACT

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CONTRIBUTORS AND SYSTEM DESCRIPTIONS

`yingfall`	**Yingfa Li**, Capital Medical University, China — **CSTS (Qwen3.5-4B):** strict extraction + Generate-Organize dual routes; deterministic graph compilation.
`ngoc`	**Ngọc Hưng Nguyễn**, University of Information Technology – VNUHCM, Vietnam — **Qwen3.5-9B:** hybrid TF-IDF + character n-grams; annotation-regime classifier.
`alan`	**Alan T. Andrea**, OptiMLX Inc., United States — **Mistral-7B LoRA:** criterion-level decomposition using 1,054 examples; CPU inference.
`mdraihan`	**Md Raihan**, Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany — **Approach not specified** in the provided table.
`kirscher`	**Tristan Kirscher**, ICube Laboratory, CNRS / University of Strasbourg, France — **Flan-T5-base:** 948 criterion-level examples; beam search; regex validation.
`ntua`	**Kyriaki Kolpetinou, Nikolas Robotis, George K. Matsopoulos**, NTUA, Greece — **Gemma-2-9B-it + PubMedBERT:** criterion-count reranking; 5-shot prompting; Q4_K_M quantization.
`harsh`	**Harsh Rajbhar**, Indian Institute of Information Technology, Design and Manufacturing, India — **LoRA Qwen2.5-1.5B:** BM25 retrieval; LoRA fine-tuning (rank 32).
`mohit`	**Mohit**, Independent Researcher, India — **Claude + MMR retrieval:** Maximal Marginal Relevance (k=4); 60-relation vocabulary.
`jason`	**Jason Karpeles**, PMG, United States — **RAG + Qwen3.5-9B:** TF-IDF retrieval; length-aware chunking; deterministic post-processing.

AI usage: ChatGPT and Manus used for proofreading, typesetting, and statistical validation. • Data: Kaggle CohortX Task 2

CohortX Task 3

Resolving Medical Conditions to ICD-10-CM Codes for Reproducible Cohort Definition

123 registered participants	**5,613** submissions	**6** validated systems	**23** test conditions	**97,441** ICD-10-CM codes	**0.438** best ensemble Avg. F1

BACKGROUND & OBJECTIVE

- EHR cohort definition requires mapping clinical conditions to ICD-10-CM codes.
- Code lists are often inconsistent or incompletely reported, producing materially different prevalence estimates.
- A role-aware benchmark evaluates structured clinical judgment—not flat code retrieval.

THREE CODE ROLES

KEEP	defines condition / cohort inclusion
ASSOCIATION	related code requiring verification
DIFF	differential diagnosis to exclude

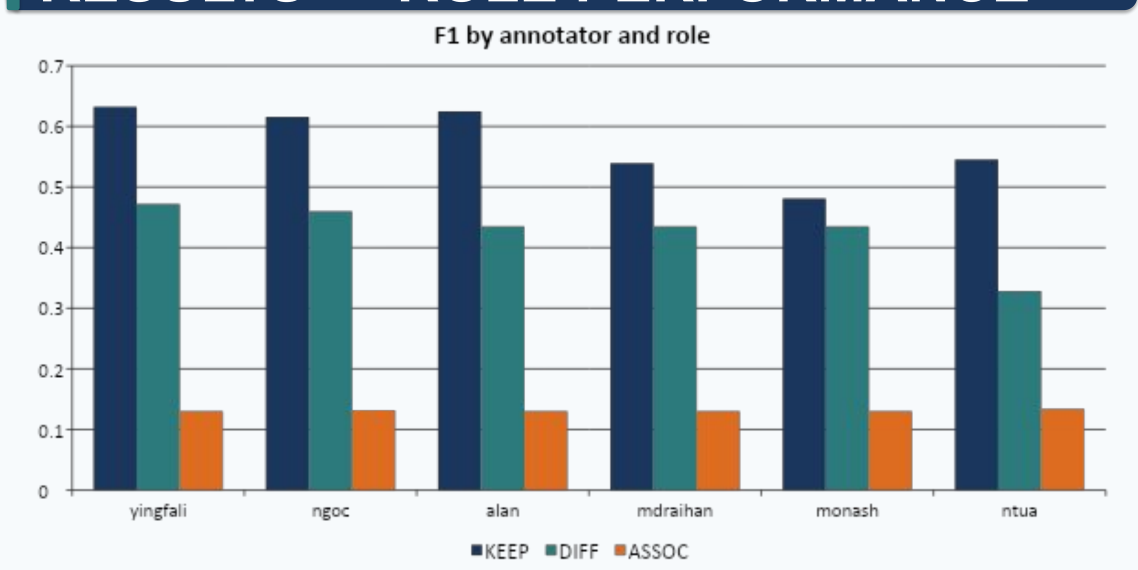
METHODS

- Evidence gathering: literature-grounded queries via Google Scholar Labs.
- Code extraction: regular expressions capturing ICD-10-CM codes and ranges.
- Resolution & annotation: mapped to MIMIC-IV dictionary; assigned KEEP / ASSOCIATION / DIFF roles.
- Training: 5 worked examples • Test: 23 evaluated conditions.
- Evaluation: mean F1 across every condition × role cell, with explicit empty-cell handling.

EVALUATION

Mean F1 = average over all condition × role cells. Equal weight to each pair; empty/empty = 1, exactly-one-empty = 0.

RESULTS — ROLE PERFORMANCE



KEY FINDINGS

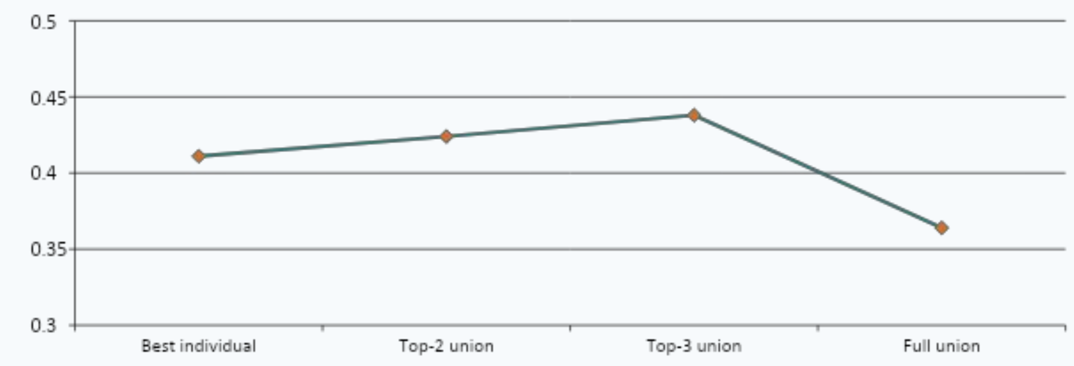
- KEEP is strongest (0.48–0.63).
- ASSOCIATION is weakest (~0.13) across all systems.
- DIFF varies more (0.33–0.47).

ERROR PATTERNS

Under-annotation dominates

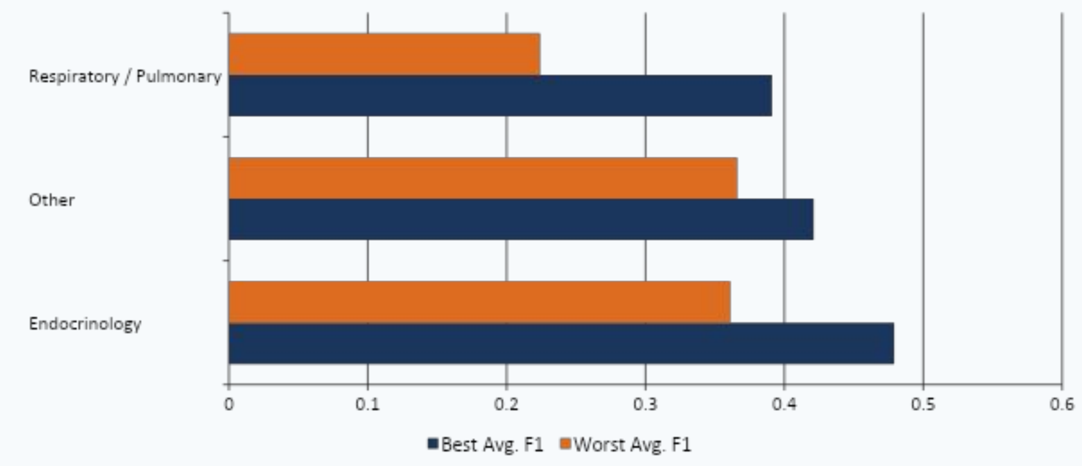
- Solution mean per case: 87.7 KEEP, 57.7 ASSOC, 65.0 DIFF.
- Most annotators produced only 35–86% of KEEP volume and near-zero ASSOC/DIFF.
- 560 solution KEEP items received zero votes.
- 99% of ASSOC items and 82% of DIFF items received zero votes.

ENSEMBLE EFFECT



Best 3-team union: 0.438 • +6.6% vs. best individual

SPECIALTY PERFORMANCE



Respiratory cases were systematically harder.

SYSTEM DESCRIPTIONS

yingfali	Qwen-assisted hierarchical retrieval with role-aware ICD expansion.	0.37995
ngoc	SapBERT + TF-IDF family ranking with role-specific classifiers.	0.37765
alan	LoRA-tuned MediPhi-Clinical with deterministic code expansion.	0.37707
mdraihan	Zero-shot BM25 + semantic retrieval with family-level selection.	0.37042
monash	KEEP-only SPLADE/SapBERT retrieval with code expansion.	0.34882
ntua	Qwen-based name generation with deterministic ICD resolution.	0.30942

DISCUSSION

- Role-aware framing exposes real reasoning: systems excel at KEEP but struggle with nuanced ASSOCIATION/DIFF distinctions.
- Ensemble fusion helps only with careful selection; adding more annotators can accumulate false positives.
- Code review is essential: 3 of 9 submitted systems were disqualified for hard-coded mappings.
- ICD coding benchmarks usually code documents; CohortX instead determines which codes define a cohort.

CONCLUSION & IMPACT

CohortX Task 3 provides a reproducible framework for evaluating condition-to-code resolution systems.

It advances computable phenotyping and biomedical informatics by making cohort-code resolution explicit, role-aware, and measurable—supporting more transparent, reproducible, and scalable cohort development.

LIMITATIONS

- Reference standard quality depends on literature and expert annotation.
- ICD-10-CM practices vary across institutions and jurisdictions.
- Only 6 validated systems; results may not generalize to all cohort-development settings.

FUTURE WORK

- Expand to additional coding systems.
- Use richer phenotype representations.
- Evaluate more complex clinical scenarios.
- Improve role-sensitive ensemble and verification strategies.

FUNDING & ACKNOWLEDGMENTS

Saudi National Institute of Health (Saudi NIH) Grant PRI01-2401-KAU19-46641119 • ICD-10 dictionary: MIMIC-IV Clinical Database Demo v2.2

CONTACT

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ntua	**Kyriaki Kolpetinou, Nikolas Robotis, George K. Matsopoulos**, School of Electrical and Computer Engineering, National Technical University of Athens, Athens, Greece

AI usage: ChatGPT and Claude used for proofreading, typesetting, and statistical validation. • Data: Kaggle CohortX Task 2 / MIMIC-IV dictionary.